

to be more similar to pretraining data in terms of n -gram counts. To encourage generalization, we can rewrite the prompt to be less similar to training data.

We implement a simple prompt optimizer based on GPT4o and the WIMBD n -gram count feedback. More specifically, we instruct GPT4o (Achiam et al., 2023) to rewrite a given task prompt at each iteration, and give the average n -gram count in the pretraining corpus of the rewritten prompt to GPT4o in the next iteration as the reward. We instruct GPT4o to maximize this reward if we want to encourage memorization, and instruct it to minimize this reward if we want to encourage generalization. Here, we show a maximization and a minimization result for TriviaQA and GSM8K respectively. We report zero-shot testing accuracy with the Pythia models and OLMo models in Table 1. The meta prompt we used to perform such optimization and the optimized task prompts are included in Appendix E.

Note that the lengths of the optimized prompts are not significantly different, while TriviaQA significantly benefits from the prompts that are more similar to the pretraining data, and GSM8K benefits from the prompts that are less similar. More sophisticated prompt optimization algorithms with more detailed distributional memorization feedback can be designed based on a similar idea. We leave the investigation of other possibilities for future work.

8 RELATED WORK

Understanding LLMs’ capabilities from training data. Most work on understanding LLMs analyzes their capabilities via synthetic experiments or small-scale studies (Arora and Goyal, 2023; Prystowski et al., 2023; Wang et al., 2024; Xie et al., 2022; Wang et al., 2023; Chan et al., 2022; Razeghi et al., 2023; Chen et al., 2024), despite the importance of scaling. Kirchenbauer et al. (2024) estimate dependencies between model capabilities and subsets of training data using kernel-based statistical evidence but rely on a small fraction of pretraining data due to computational constraints. To address this, Elazar et al. (2024) introduce WIMBD, a system for efficient n -gram retrieval over massive datasets, while Merrill et al. (2024) develop an unbounded n -gram search method and find larger LLMs generate fewer novel n -grams. Shaib et al. (2024) demonstrate how syntactic templates in training data influence LLM outputs, and Liu et al. (2024) propose ∞ -gram language models to estimate text corpora distributions, focusing on local dependencies rather than complex capabilities.

In this work, we analyze the origins of LLMs’ zero-shot capabilities by leveraging WIMBD and ∞ -gram frameworks for full-scale pretraining data exploration.

Memorization vs. generalization. LLM memorization, defined as exact recall of training data, has been extensively studied, including memorization of rare or private data (Zhang et al., 2023), test set contamination (Jiang et al., 2024), and higher prevalence of verbatim recall in larger models (Carlini et al., 2022). Hartmann et al. (2023) provides a comprehensive summarization of different types of LLM memorization. The relationship between memorization and generalization is explored in Feldman (2020), showing that memorization can improve generalization. Extensions to this work quantify memorization via performance differences when specific training examples are included or excluded (Feldman and Zhang, 2020; Zhang et al., 2023). However, this approach is infeasible for large-scale LLMs due to retraining requirements. We propose a scalable definition of distributional memorization using n -gram counts to enable efficient large-scale analysis.⁵

9 CONCLUSION

In this paper, we introduce the **task-gram language model**, a scalable approach for modeling the task-relevant distribution of pretraining data. We trace the capabilities of LLMs back to this data by defining **distributional memorization**, measured through the Spearman correlation between task-gram LM probabilities and LLM probabilities. Through extensive experiments using Pythia models across four distinct tasks, we find that LLMs tend to memorize more when engaged in simpler, knowledge-intensive tasks, while they generalize more in harder, reasoning-intensive tasks. Our analysis provides a comprehensive examination of the origins of LLM capabilities and offers a scalable framework for investigating the fine-grained task-relevant characteristics of pretraining corpora.

⁵A More detailed discussion of related work can be found in Appendix C.

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